LED House — Electrical Shop

"Where every home finds its light."

Software Analysis & Design — System Documentation

Albania · 2025

Section	Title
01	Photo Ethnography
02	Onion Diagram — Stakeholder Map
03	User Scenarios
04	User Story Cards
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01 — Photo Ethnography

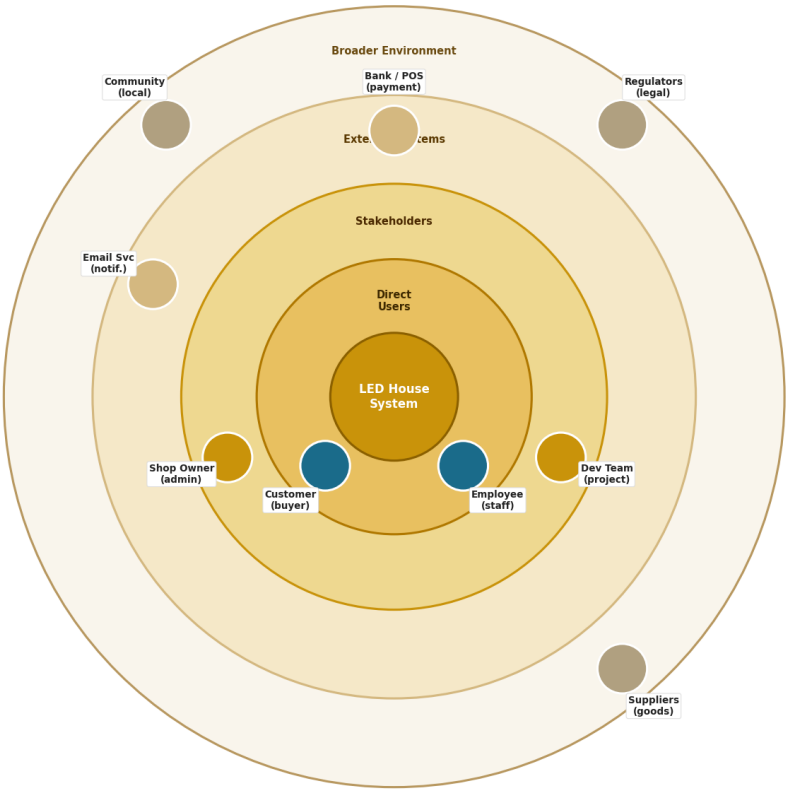
Visual documentation of how LED House currently operates as a physical shop. Each scene identifies a real-world workflow, pain point, or interaction that the proposed system must address.

Key findings: The shop operates entirely offline. Main pain points are: (1) no after-hours product discovery; (2) manual stock checking by walking to storage; (3) paper order records with no backup; (4) handwritten prices without a catalogue. The LED House system targets all four.

02 — Onion Diagram: Stakeholder Map

The onion diagram maps stakeholders by their proximity to the system core. Inner layers interact directly with the system; outer layers are impacted indirectly.

Onion Diagram — LED House Stakeholders



Layer	Who	Relationship
System Core	LED House platform	The software being built
Direct Users	Customer, Employee	Daily hands-on interaction
Stakeholders	Shop Owner, Dev Team	Decision-making & development
External Systems	Bank POS, Email Service	Integration points
Broader Environment	Regulators, Suppliers, Community	Indirect influence

03 — User Scenarios

Narrative descriptions of how different stakeholders interact with the system in real-world context. Each scenario identifies a cluster of system requirements.

User Scenarios — LED House

Scenario 1: Artan — Home Renovator

Customer | occasional buyer | non-technical

"I need LED strips for my bathroom — I want to order at 9 PM when the shop is closed."

1. Opens LED House website on mobile at 21:00
2. Types "LED strip warm white" in search bar
3. Filters by price (< €20) and In Stock
4. Reads product specs and description
5. Adds 2× 5m strip to cart → Checkout
6. Pays via bank POS link
7. Receives email confirmation
8. Collects order at shop next morning

✓ Purchase completed outside business hours — previously impossible.

Scenario 2: Elira — Electrician

Customer | frequent buyer | technical user

"I need to check if 16A type-B breakers are in stock before driving across town."

1. Searches "circuit breaker 16A" in catalogue
2. Views product page: 4 units in stock
3. Places a click & collect reservation
4. Drives to the shop — items already set aside
5. Pays at counter via POS terminal

✓ Saves 40-min round trip. Stock confirmed digitally in 2 minutes.

Scenario 3: Gent — Shop Owner

Admin | manages everything | strategic user

"Monday morning: I need to see weekend orders, update stock, and adjust prices."

1. Logs into admin dashboard
2. Reviews 7 new weekend orders
3. Marks 2 orders as "Collected"
4. New stock arrived → updates quantities for 5 products
5. Edits prices for 3 LED bulb variants
6. Checks low-stock alerts (2 items flagged)

✓ Full weekly operations done in 15 min — was 60+ min manually.

Scenario 4: Drita — Store Employee

Staff | daily ops | order fulfilment

"A customer calls — can I check stock and create an order without asking the owner?"

1. Logs into staff dashboard (limited access)
2. Searches catalogue: 23m of 2.5mm² cable in stock
3. Confirms availability to customer over phone
4. Creates manual order with customer details
5. Marks as "Awaiting Payment"
6. Sets aside items and attaches order slip

✓ Staff handles orders independently — owner not interrupted.

04 — User Story Cards

Formal user stories using the template: *As a [role], I want [feature], so that [benefit].* Each card includes acceptance criteria and priority level.

User Story Cards — LED House

US-01 HIGH

Browse product catalogue

As a customer,
I want to browse products by category,
so that I can discover what the shop offers without visiting

Acceptance criteria:

- ∨ Products grouped by category
- ∨ Shows name, price, photo, stock
- ∨ Loads in < 3s on mobile
- ∨ Works on all screen sizes

Catalogue

US-02 HIGH

Search & filter products

As a customer,
I want to search by keyword and filter by price,
so that I find my product fast.

Acceptance criteria:

- ∨ Keyword matches name/description
- ∨ Filters: price, category, in-stock
- ∨ Results update dynamically
- ∨ Empty state with suggestions

Search

US-03 HIGH

Add to cart & place order

As a customer,
I want to add products and complete a purchase online,
so that I can buy any time.

Acceptance criteria:

- ∨ Cart persists during session
- ∨ Checkout needs name & phone
- ∨ Payment via bank POS redirect
- ∨ Stock reduced on confirmed order

Cart & Payments

US-04 HIGH

Manage products & inventory

As the shop owner,
I want to add, edit, delete products with price & stock,
so that the catalogue is accurate.

Acceptance criteria:

- ∨ Secure admin login required
- ∨ Create: name, category, price, photo
- ∨ Changes live within 5 seconds
- ∨ Low-stock alert at threshold

Admin / Inventory

US-05 HIGH

View & manage orders

As the shop owner,
I want to see all orders and update their status,
so that I track sales daily.

Acceptance criteria:

- ∨ List: ID, customer, items, total, status
- ∨ Filter by status and date
- ∨ Update: Pending/Ready/Collected
- ∨ Order history kept 12+ months

Orders

US-06 MED

Receive order confirmation

As a customer,
I want to receive a confirmation after placing an order,
so that I have proof of purchase.

Acceptance criteria:

- ∨ Email sent within 2 minutes
- ∨ Shows order ID, items, total
- ∨ On-screen success page shown
- ∨ Includes estimated pickup info

Notifications

US-07 MED

Secure admin & staff login

As the shop owner,
I want to control who accesses what features,
so that staff can work safely.

Acceptance criteria:

- ∨ Roles: Admin (full) / Staff (limited)
- ∨ Login with email + password
- ∨ Session expires on inactivity
- ∨ Staff cannot edit prices/financials

Security

US-08 MED

Low-stock alerts

As the shop owner,
I want to be alerted when stock falls below a threshold,
so that I reorder in time.

Acceptance criteria:

- ∨ Configurable threshold per product
- ∨ Alert shown in admin dashboard
- ∨ Email notification to owner
- ∨ Alert cleared on stock update

Stock Mgmt

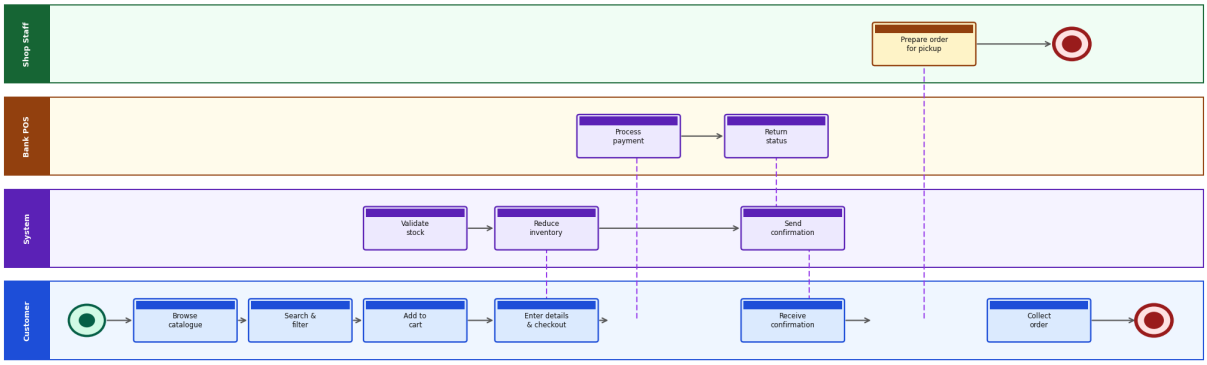
05 — BPMN Process Diagrams

Business Process Model and Notation (BPMN 2.0) diagrams for the five core workflows. Pools represent organisational participants; dashed arrows indicate message flows between pools (cross-organisational). Solid arrows are sequence flows within a pool.

BPMN 1: Customer Online Purchase

End-to-end flow from product discovery through payment and order pickup. Spans Customer, System, Bank POS, and Shop Staff pools.

BPMN 1 — Customer Online Purchase Flow



BPMN 2: Admin Product Management

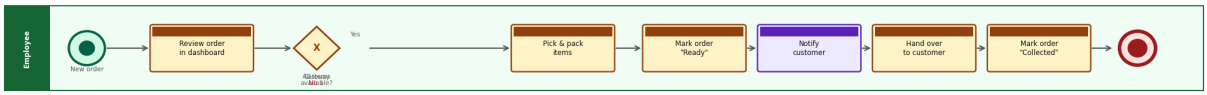
Shop owner adds a new product or updates existing stock and pricing through the admin dashboard.

BPMN 2 — Admin Product Management Flow

BPMN 3: Order Fulfilment

From a new order notification to the customer collecting it at the counter.

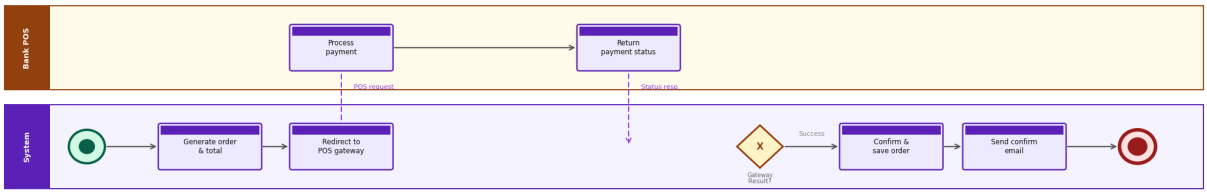
BPMN 3 — Order Fulfilment Flow (Employee)



BPMN 4: Payment Processing

How the LED House system interacts with the external bank POS for payment authorisation.

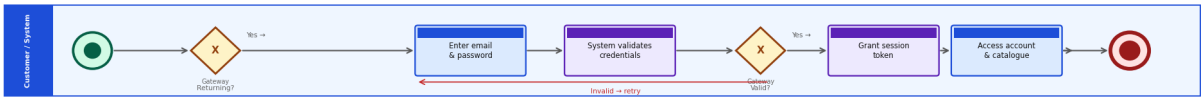
BPMN 4 — Payment Processing Flow



BPMN 5: Customer Registration & Login

How a new or returning customer authenticates before placing an order, including retry on invalid credentials.

BPMN 5 — Customer Registration & Login Flow



BPMN Notation Legend

Symbol	Name	Meaning
● (green thin border)	Start Event	Process begins here
● (red thick border)	End Event	Process terminates here
Blue-topped box	User Task	Action performed by a human
Purple-topped box	Service Task	Automated system action
Yellow-topped box	Manual Task	Physical or offline action
Diamond with X	Exclusive Gateway	Only one path continues (XOR)
Solid arrow →	Sequence Flow	Order of activities within a pool
Dashed purple arrow	Message Flow	Communication between pools

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